

Array - Two Dimension

1. Write a menu driven C++ program to do the following operation on two-dimensional array A of size $m \times n$. You should use user-defined functions which accept 2-D array A, and its size m and n as arguments. The options are:

- To input elements into matrix of size $m \times n$
- To display elements of matrix of size $m \times n$
- Sum of all elements of matrix of size $m \times n$
- To display row-wise sum of matrix of size $m \times n$
- To display column-wise sum of matrix of size $m \times n$
- To create transpose of matrix B of size $n \times m$

2. Write user defined functions for square matrix to calculate

1. Left diagonal sum
2. Right diagonal sum

3. Write a user-defined function in C++ to display the multiplication of row element of two-dimensional array A [4][6] containing integer.

4. Write a user defined function named Upper half () which takes a two dimensional array A, with size N rows and N columns as argument and prints the upper half of the array.

e.g.,

2 3 1 5 0	2 3 1 5 0
7 1 5 3 1	1 5 3 1
2 5 7 8 1	1 7 8
0 1 5 0 1	0 1
3 4 9 1 5	5

Output will be:

5. Write a function in C++ which accepts a 2D array of integers and its size as arguments and displays the elements of middle row and the elements of middle column.

[Assuming the 2D Array to be a square matrix with odd dimension i.e., 3×3 , 5×5 , 7×7 etc...]

Example, if the array contents is

3 5 4
7 6 9
2 1 8

Output through the function should be:

Middle Row : 7 6 9

Middle column : 5 6 1

- 6.** Write a program to add two array A and B of size $m \times n$.
- 7.** Write a program to multiply array A and B of order $N \times L$ and $L \times M$